

✦ PRODUCTION DEPLOYMENT BLUEPRINT • 100% FREE TIER

Live Cloud Deployment Manual

Comprehensive step-by-step instructions for deploying **IntelliMeet** to public HTTPS/WSS endpoints using Vercel, Render, and MongoDB Atlas with zero hosting costs and full WebRTC signaling support.

Architecture: Jamstack + Cloud Containers

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SSL/TLS: Automated Let's Encrypt

Document ID: ZIDIO-DEPLOY-05

🏗️ High-Availability Cloud Architecture Blueprint



🚀 Step-by-Step Deployment Instructions

1 Database Setup (MongoDB Atlas Managed Free Tier)

1. Sign in to mongodb.com/cloud/atlas and click **Create Deployment** → Select **M0 Free Cluster**.
2. Choose provider region (e.g., AWS / Mumbai or Frankfurt) closest to primary users.
3. Under **Database Access**, create an admin user (e.g., `intellimeet_admin`) with a strong password.
4. Under **Network Access**, add IP `0.0.0.0/0` (Allow Access from Anywhere) to permit Render containers to connect.
5. Click **Connect** → **Drivers** and copy your connection string:

```
mongodb+srv://intellimeet_admin:<password>@cluster0.abcde.mongodb.net/intellimeet?retryWrites=true&w=majority
```

2 Backend Service Deployment (Render.com Web Service)

1. Sign in to render.com and click **New +** → **Web Service** → Connect repository `harsadash/intelliMeet`.

2. Configure settings: **Root Directory:** server | **Runtime:** Node | **Build:** npm install && npm run build | **Start:** npm start.
3. Under **Environment Variables**, add NODE_ENV=production, MONGO_URI, JWT_SECRET, JWT_REFRESH_SECRET, and CLIENT_URL.
4. Click **Create Web Service**. Render provisions the container and assigns: <https://intellimeet-api.onrender.com>.

Step-by-Step Deployment Instructions (Contd.)

3 Frontend Client Deployment (Vercel Free Tier)

1. Sign in to vercel.com with GitHub and click **Add New...** → **Project**.
2. Import repository `intelliMeet`. Select Framework Preset: **Vite**.
3. Set **Root Directory**: `client` | **Build Command**: `npm run build` | **Output Directory**: `dist`.
4. Under **Environment Variables**, configure:

```
VITE_API_URL=https://intellimeet-api.onrender.com
VITE_SOCKET_URL=https://intellimeet-api.onrender.com
```

5. Click **Deploy**. Vercel compiles React assets and assigns: `https://intellimeet-app.vercel.app`.

4 Bidirectional Linkage & CORS Origin Synchronization

1. Navigate back to Render Dashboard → `intellimeet-api` → **Environment**.
2. Update `CLIENT_URL` to match your production Vercel domain: `https://intellimeet-app.vercel.app`.
3. Click **Save Changes**. Render automatically cycles and redeploys the service with the new CORS origin.

Production Environment Configuration Matrix

Detailed reference of all production variables required across backend and frontend environments:

Variable Key	Scope	Description & Requirement Standard	Default / Sample
NODE_ENV	BACKEND	Enables Express production optimizations, secure cookies, and disabled stack traces.	production
PORT	BACKEND	Web server listen port. Cloud platforms (Render/Railway) inject this automatically.	10000 / 5000
MONGO_URI	BACKEND	Connection string for MongoDB Atlas cluster (with credentials and DB name).	mongodb+srv://...
JWT_SECRET	BACKEND	Cryptographic secret key for signing short-lived access JWT tokens (15m expiration).	32+ Char Random
JWT_REFRESH_SECRET	BACKEND	Independent secret key for signing long-lived rotatable refresh tokens (7d).	32+ Char Random
CLIENT_URL	BACKEND	Authorized frontend domain for CORS whitelist and Socket.io handshake validation.	https://...vercel.app
OPENAI_API_KEY	OPTIONAL	API key for GPT-4o-mini synthesis. If omitted, offline heuristic synthesizer runs.	sk-proj-...
VITE_API_URL	FRONTEND	Target backend URL for Axios HTTP requests (/api/auth, /api/meetings, /api/tasks).	https://...onrender.com
VITE_SOCKET_URL	FRONTEND	Target backend WebSocket gateway for Socket.io signaling & in-room chat.	https://...onrender.com

Production File Templates

Backend Template: server/.env.production

```
NODE_ENV=production
PORT=5000
MONGO_URI=mongodb+srv://admin:pass@cluster0.xyz.mongodb.net/intellimeet?
retryWrites=true&w=majority
JWT_SECRET=super_secret_jwt_access_token_key_2026
JWT_REFRESH_SECRET=super_secret_jwt_refresh_token_key_2026
JWT_EXPIRE=15m
JWT_REFRESH_EXPIRE=7d
CLIENT_URL=https://intellimeet-app.vercel.app
OPENAI_API_KEY=sk-proj-your_key_here_or_leave_empty
```

Frontend Template:
client/.env.production

```
# Production REST API Endpoint
VITE_API_URL=https://intellimeet-
api.onrender.com

# Production WebSocket Gateway
VITE_SOCKET_URL=https://intellimeet-
api.onrender.com

# Build Mode Indicator
VITE_APP_ENV=production
```

Security Warning: Never commit filled .env.production files with real secrets to public GitHub repositories. Inject values directly via the Vercel & Render dashboards.

Live Verification Matrix (Post-Deployment Audit)

- ✓ HTTPS SSL Validation: Frontend URL loads with a valid security padlock icon; zero mixed-content warnings.
- ✓ WSS Protocol Upgrade: Socket.io signaling establishes connection over wss:// without HTTP polling fallback.
- ✓ CORS Whitelist: Inspecting DevTools Network panel reveals zero Access-Control-Allow-Origin errors.
- ✓ Database Read/Write: Logging into employee@intellimeet.io retrieves meetings and saves new tasks to Atlas.
- ✓ Multi-Peer WebRTC: Two separate browser windows join /meeting/demo-room-101 and stream video peer-to-peer.
- ✓ AI Post-Meeting Summary: Clicking "Generate Summary" delivers structured takeaways on the live domain.

Free-Tier Operations & Performance Tuning

Cold-Start Mitigation

Render free web services spin down after 15 minutes of inactivity. Initial requests may experience a 30–45 second wake-up delay. A free health-check ping (e.g., UptimeRobot pinging /api/health every 10 minutes) keeps the container active during evaluation windows.

STUN / TURN Relay Traversal

Default STUN configuration utilizes Google's public STUN servers (stun:stun.l.google.com:19302). For symmetric NAT enterprise firewalls, adding a free Metered.ca TURN server credential ensures 100% video connectivity across restricted networks.

LEAD DEVOPS & FULL-STACK ARCHITECT

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STATUS: PRODUCTION CERTIFIED

Verified: March 2026